PDF Conversion Guide

This document outlines the standardized process for converting Markdown (`.md`) files to PDF in the Mitsue Village Project AI Data Center.

Why This Method?

We use a custom Node.js script (`convert_md_to_pdf.js`) powered by Puppeteer and the `markdown-pdf` extension. This method was chosen over alternatives (like Pandoc + LaTeX) because it provides:

1. **Superior Visual Fidelity**: Retains custom CSS styling, fonts, and Typora theme compatibility.
2. **Reliable CJK (Japanese) Text Wrapping**: Solves the critical issue of Japanese text overflowing table boundaries, which is a known limitation in standard LaTeX `tabular` environments.

Prerequisites

- Node.js installed on the system.
- The `yzane.markdown-pdf` extension installed (located at `~/.antigravity/extensions/yzane.markdown-pdf-1.5.0-universal`).

The Japanese Table Fix (Critical)

The script includes a `PRINT_OVERRIDE` CSS constant that enforces strict line-breaking rules specifically for table cells. This prevents long Japanese strings from breaking the layout. The critical CSS rules applied to `table th, table td` are:

```
line-break: anywhere !important;
word-wrap: break-word !important;
overflow-wrap: break-word !important;
word-break: break-all !important;
```

Mermaid Diagram Support

The script natively supports rendering Mermaid.js diagrams (e.g., flowcharts, sequence diagrams) from ````mermaid` code blocks.

- It uses a local copy of `mermaid.min.js` (v9.4.3) to avoid browser security restrictions on `file://` protocol CDN loading.
- Version 9.x is specifically used because newer versions (v10+) require `structuredClone`, which is not supported by the older Chromium engine bundled with the extension.
- The script intercepts Mermaid code blocks and wraps them in `<div class="mermaid">`, then waits 3 seconds for the SVG rendering to complete before generating the PDF.

Standard Workflow

Step 1: Update Document Headers

Before converting, always ensure the document's version and date are up to date. Run the header update script:

```
python3 update_doc_headers.py
```

This script automatically increments the version number and sets the "Last modified" date to the current date (e.g., `2026-06-05`) in the HTML header of all `.md` files.

Step 2: Convert to PDF

Run the conversion script. You can target specific files or convert all Markdown files in the directory.

Convert specific files:

```
node convert_md_to_pdf.js file1.md file2.md
```

Convert all Markdown files in the directory:

```
node convert_md_to_pdf.js --all
```

Troubleshooting

- **Japanese text still overflowing?** Verify that `convert_md_to_pdf.js` contains the `PRINT_OVERRIDE` CSS rules mentioned above. Do not remove or alter these rules.
- **Script fails to run?** Ensure Node.js is installed and the `markdown-pdf` extension path is correct in the script's configuration.
- **Need smaller file sizes?** A fallback script (`convert_md_to_pdf_pandoc.py`) exists using Pandoc + `xelatex`, but it may lose custom CSS styling and requires manual `tabularx` configuration for Japanese tables. Use only if file size is a strict constraint and visual fidelity is secondary.

Last updated: 2026-06-05